

Upper limit effects in DPs and the interdependence of worlds and times¹

Paul MEISENBICHLER — *Massachusetts Institute of Technology (MIT)*

Abstract. The distribution of upper limit effects suggests that an expression’s evaluation time cannot be bound independently from its evaluation world. Specifically, even in cases where worlds and times can be non-locally bound (which implies the availability of mechanisms with object-language quantification or equivalent power), both coordinates must be bound by the same (non-local) antecedent. From this, I conclude that the mechanism manipulating worlds and times in natural language is less expressive than a mechanism that manipulates independent object-language variables that range over worlds and times. At most, the mechanism that binds worlds and times can express binding of evaluation indices (i.e. world-time pairs).

Keywords: de re, de dicto, upper limit effects, world variables, time variables.

1. Introduction

1.1. Non-local binding of worlds and times

The component of natural language (NL) that expresses temporal and modal thoughts (i.e. propositions that are not about the *actual here and now*) can be described by grammars that quantify over times and possible worlds. Several decades of research into modal and temporal quantification have revealed a great deal about the kinds of syntactic and semantic mechanisms that NL employs in this domain. One of the main discoveries is that quantification over worlds and times can establish *non-local* binding dependencies, which suggests that it relies on mechanisms that are as expressive as *object language quantification*.² Consider (1) and (2).

- (1) a. Scenario: *There are four millionaires at Charlotte’s party: Albert, Andrew, Jerry, and Henry. Tony doesn’t know that they are millionaires, but he quickly forms the opinion that all four of them are unpleasant and arrogant. When Charlotte tells him a rumor that there is a person at the party who has dated all four of them, he quickly rejects this possibility: “It is unimaginable to me that there would be a person with such bad taste in men. I can maybe see a guest dating one or two of them and regretting it, but all four? Impossible”. Charlotte reports what Tony thinks about the rumor by uttering (1b).*
- b. Tony believes that no guest has dated *every millionaire at the party*.
- (2) a. Scenario: *As an adult, James faces multiple charges of corruption. When the possibility of conviction comes up, Bill is skeptical. “Someone like James – a wealthy man from a well-connected, aristocratic family – will never face real consequences. He’s friends with nearly everyone who holds power or money in this country. Half*

¹I would like to thank Amir Anvari, Tamari Berulava, Danny Fox, Martin Hackl, Viola Schmitt and the participants of *NELS 55* and *Sinn und Bedeutung 30* for helpful comments and suggestions. All mistakes are my own.

²See Fodor (1970), Kamp (1971), Vlach (1973), Enç (1981, 1986), Bäuerle (1983), Hoeksema (1984), Heim (1985, 1991), Cresswell (1990), Abusch (1994), Farkas (1994), Percus (2000), Kusumoto (2005), von Stechow (2011), Schwarz (2012), Elbourne (2013), among many other works that have provided arguments for a conclusion along these lines.

the federal judges who might preside over his case are people he's known since high school, back when they were spoiled teenagers drinking beer together at a party almost every week."

- b. Whenever James went to a party in high school, he drank beer with *a federal judge*.

Example (1) shows that the evaluation world of a DP is not always determined locally by the syntactically closest world quantifier. The evaluation world of *every millionaire at the party* is the actual world even though the DP's quantificational force is evaluated inside the attitude context, forcing it to scope below *believe* at LF. Tony does not rule out that a guest might have dated one or two of the (actual) millionaire guests. He only objects to the possibility that anyone could have dated *all* of them, meaning *every millionaire at the party* scopes below the negative existential *no guest*. Since *no guest* clearly scopes below the attitude verb, transitivity dictates that the universal quantifier must do so as well.³ In (2), there is no specific (current) judge that James drank beer with at every high school party, so *a federal judge* must scope below the *whenever*-clause (the time quantifier which universally quantifies over the party occasions). Still, the indefinite is clearly evaluated at the utterance time and not relative to the "local" party times (in high school, the relevant individuals had not been judges yet).^{4 5}

It is worth emphasizing that non-local evaluations of the sort we find in (1) and (2) are not exclusive to situations where the DP takes the world and time of *utterance* as the "non-local" antecedent. If that were the case, then all we would have shown is that the utterance context is

³The non-existent reading, where the negative existential takes inverse scope over the attitude verb, would require that there are worlds compatible with Tony's beliefs in which a guest did in fact date all four millionaires. This means Tony would consider the existence of such a person *possible*, which is not the case in the given context and it is in fact not a scenario which this sentence can describe.

⁴The availability of non-local world and time binding a.) generalizes to other determiners/quantifiers than the ones in these examples and b.) generalizes to other strategies of fixing an expression's LF-position. Hoeksema (1984) and Heim (1985) show that the world of a degree clause α can be bound by an antecedent that is non-local to the position that α has to occupy because a structurally lower quantifier binds a pronoun inside α . Kusumoto (2005) and Elliott (2023) show that world and time binding can be dissociated from the position that an expression needs to occupy for *NPI-licensing*. Cresswell (1990), Abusch (1994) and Elliott (2023) give examples where the relevant expressions depend on a world/time that is non-local to the (scope) islands these expressions are trapped in (*if*-clauses).

⁵(1) and (2) make use of slightly different flavors of "scope trapping": while (2) gives a scenario that directly controls for the quantificational scope of the target DP relative to the intensional operator (creating examples known as *Fodor's third reading* in the literature, see Fodor, 1970), (1) does this only indirectly by independently fixing the scope of both the DP and the intensional operator relative to a third expression (such cases are known as *Bäuerle sentences*, see Bäuerle, 1983). Importantly, we can create Fodor-type cases for world binding (see (3)) and Bäuerle-type cases for time-binding. (i) gives a time binding example that more closely resembles (1) (see Meisenbichler, 2025).

- (i) a. Scenario: *In the early 2010s, a psychologist conducted a series of experiments on online dating. Participants were shown photographs of students from a nearby university through a fake dating platform. For each photograph, they had to decide – based solely on appearance – whether they would accept or reject the person for a date. Among the students who volunteered their pictures were Bob, Lisa, and Tom, all of whom would later become professors in the university's mathematics department. In 2025, the psychologist's computer was hacked, and the experimental data were made public. A member of the mathematics department noticed a striking regularity: In every experiment, exactly thirty participants rejected at least one of Bob, Lisa, or Tom, and no participant rejected all three. Upon discovering this, he remarked:* (ib).
- b. Whenever the psychologist ran an online-dating experiment, exactly thirty participants rejected *a professor from our department*.

special and that the world and time coordinates of the context are always accessible as potential sources of evaluation. DPs can also enter into genuine quantificational dependencies with higher modal and temporal quantifiers even if they are separated from them by structurally intervening operators. Consider (3), which involves an indefinite DP (*a German*) in the scope of world quantifiers. In contrast to the classic example, the DP is embedded under more than one intensional operator (*tell* and *think*) to demonstrate the existence of non-local world binding that goes beyond the indexical (= utterance world-dependent) reading. The indefinite takes lowest scope: Nina makes clear that her brother didn't have any specific individual in mind. His reported suspicion is merely that *one* of the neighbors is dating Nina, without this suspicion being directed at any particular neighbor. Crucially, Nina also makes clear that the descriptive content of *a German* is not her brother's description. Her brother would have never believed that she dated someone who is German. Finally, the description is also not true in the actual world, since the speaker knows that the relevant individuals are Austrians. What we have here is a situation in which the quantificational force of the DP *a German* is evaluated in the scope of the lower world quantifier *think*, but its world coordinate is bound by the higher world quantifier *claim*.

- (3) a. Scenario: *Peter and Fritz are Austrians who share a common grievance: their non-Austrian friends' chronic inability to distinguish between Austrians and Germans. The latest offender is their friend Nina, who recently recounted a misunderstanding with her brother. While visiting, her brother noticed her close rapport with her neighbors – Hans, Max, and Ludwig – and suspected she was secretly dating one of them. Nina believes her neighbors are German and that her brother mistook them for Italians. Nina told Peter: "You see, he wouldn't have suspected a thing if he had known they were German. Everyone knows I only date Italians!" Peter doesn't correct her mistake and later tells the story to Fritz to illustrate Nina's ignorance:*
- b. Nina told me that her brother thought that she was dating *a German*.

To appreciate (5), we must first recognize that attitude verbs (can) quantify over times in addition to worlds, meaning they are potential binders of the time arguments taken by expressions in their scope (see Heim, 1994; Abusch, 1997; von Stechow, 2004). This becomes clear once we are in a context in which the internal "now" of an attitude holder diverges from the attitude holder's actual time (the time specified by the tense marking on the attitude verb). A mistaken time of an attitude holder (the attitude time) is a potential antecedent for a DP's evaluation time: in (4), it is crucially 1970 and not 2026 (the utterance time) at which Gina's husband was 20 years old.⁶

- (4) a. Scenario: *Gina has dementia. Today, she believed that she was on a beach in Italy in the 1970s talking to her husband at a time when they were both 20 years old.*
- b. This morning, Gina thought that she was talking to *a 20-year-old* at breakfast.

We can now combine an attitude verb that quantifies over the attitude time of a confused attitude holder with a higher temporal quantifier that binds the actual time of the attitude event. What

⁶At this point, I am not making a clear distinction between the *evaluation time* of the DP and the *property time* of the noun (the former being the "variable-like" reference time that can be non-locally bound, the latter being the time throughout which the nominal property holds of the individuals in its extension). Later I will make the distinction, but here it doesn't matter because *20-year-old* happens to be a noun that imposes an overlap-requirement between its property time and the evaluation time (see O'Leary, 2022).

we want to test is whether a DP can be bound by the higher quantifier even if it scopes below the attitude verb. The answer is yes: in (5), the indefinite DP *a 20-year-old* needs to scope below the attitude verb. At any given test time, Susan is unsure which of the individuals in the pictures is her husband – the individual co-varies with the *worlds* that the attitude verb quantifies over. Crucially, the evaluation time of the nominal predicate cannot be determined by the attitude time because the attitude time is always 1990 (some of the “20-year-olds” are not just younger than 20 at that time but had not even been born then). The evaluation time is also not the utterance time since the relevant individuals are all older than 20 now. Instead, the evaluation time seems to be *bound* by the (actual) times picked out by the non-local *whenever*-clause.

- (5) a. Scenario: *Susan suffers from a rare mental disorder that leaves her convinced she is permanently living in the year 1990. Since the early 1990s, she has been evaluated by a specialist every five years. During these evaluations, the doctor asks her a range of questions, including the current year – which she unfailingly identifies as 1990 – as well as personal details about her life. When asked whether she is married, Susan always responds that she is unsure. She then shows the doctor three photographs of her nurses, all of whom are twenty-year-old men that are in training at the time of each examination (it is always the 20-year-old trainees that are hired to care for Susan). Pointing to the photos, she says, “I don’t know which one exactly, but I must be married to one of them. They’re always around.” When asked what she knows about the individuals in the pictures, she always comments on their age: “They look dreadfully old to me. They must be older than 30.”*
- b. Whenever the doctor evaluated Susan’s condition over the years, she believed that she was married to *a 20-year-old*.

1.2. World and time variables cannot be independent

Early attempts to develop intensional logics for natural language (e.g. Montague, 1973; and other theories inspired by classic modal logic) relied on systems where the value of an expression’s world or time coordinate is always determined by the most local modal and temporal quantifier (e.g. the most local tense, the most local attitude verb, and so on). The possibility of (unbounded) non-local binding of worlds and times showed that these purely local mechanisms are not expressive enough. What seems to be needed is a system that can a.) keep track of a (potentially infinite) sequence of worlds and times introduced by higher operators and b.) refer back to any world or time on that sequence over an arbitrary structural distance. A system that meets these requirements would for example be one that represents world- and time-arguments as object language variables in the syntax.⁷ To illustrate, if we assume that a DP’s world ar-

⁷A point that is always worth emphasizing is that the existence of non-local binding does not need to be understood as an argument for the existence of syntactic variables of the relevant sort. The argument is more nuanced: a mechanism that employs object language variables is formally expressive enough to capture non-local binding dependencies, whereas a mechanism relying on local parameter shifting is not. But there are other mechanisms whose expressive power is provably equivalent to object language quantification, but that *do not* make use of object language variables (e.g. Cresswell, 1990). What non-local binding thus shows is the following: whatever mechanism NL employs in these cases, it seems to belong formally into a class of mechanisms that have a particular expressive power. Object language quantification happens to be a natural example from this class. The conclusion that object language quantification is in fact the specific mechanism from this class that comes closest to the actual NL-mechanism requires additional argumentation, e.g. by drawing on parallels to other domains where one might

gument is syntactically saturated by a silent variable, then this variable can be co-indexed with any structurally higher world quantifier. (6a) gives a schematic LF that can derive the non-local world dependency in (3).⁸ The LF in (6b) shows how explicit time variables would allow non-local binding of the kind we found in (5).

- (6) a. $\lambda w_0 \dots \text{tell } \lambda w_4 \dots \text{think } \lambda w_6 \dots [\text{a German } w_4]$
 b. $\lambda t_0 \dots \text{whenever}_{\text{the doctor evaluated...}} \lambda t_4 \dots \text{believe } \lambda t_6 \dots [\text{a 20-year-old } t_4]$

Although the logic of the argument is sound, I argue that the two mechanisms we end up with (one manipulating *world variables*, the other manipulating *time variables*) are too powerful once they are combined into the same intensional system. Specifically, I argue that NL does not generate readings that would require the power of *independent* object language binding of worlds and times. These are readings of the kind that would be generated by the following LF (which just combines the key elements from (6a) and (6b)):

- (7) $\lambda w_0 \lambda t_0 \dots Op' \lambda w_4 \lambda t_4 \dots Op'' \lambda w_6 \lambda t_6 \dots [\alpha \dots w_6 t_4]$

Instead, the world- and time-coordinate of an expression always behave as though they depend *on the same antecedent*. In other words: if the “world variable” of an expression α is (locally or non-locally) bound by some intensional operator β , then α ’s “time variable” must be bound by β too. This is equivalent to saying that the mechanism that NL employs when it quantifies (non-locally) over worlds and times is only as expressive as object language quantification over *index variables* (i.e. variables that range over *evaluation indices*, which are complex objects with several contextual coordinates but can for our purposes simply be conceived of as world-time pairs).⁹ An LF like (8) – where i denotes a world-time pair – describes a substantially less expressive mechanism than (7). I argue that there is no evidence that NL has access to a mechanism that is more powerful than (8).

- (8) $\lambda i_0 \dots Op' \lambda i_4 \dots Op'' \lambda i_6 \dots [\alpha \dots i_4]$

The argument goes as follows:

- (9) The argument (in a nutshell)
 a. The *property time* of a nominal predicate (the time throughout which the individuals in the extension have the nominal property) must be distinguished from the noun’s *evaluation time* (which is the time that can be non-locally bound in a DP).

be convinced that object language variables are employed (i.e. in the domain of individual pronouns). I remain agnostic whether non-local binding in NL makes use of object language variables or an equivalent mechanism, but I will use variables because they are clear illustrations of the relevant *class* of mechanisms I have in mind. This is to say, if I argue that a mechanism with *index variables* is more adequate than mechanisms with independent variables for worlds and times, then this should again be understood as an argument about the formal nature of the underlying NL-mechanism and not as an argument about the existence of the *particular* mechanism that is used for illustration (i.e. object language index variables).

⁸Assuming standard assumptions of compositional interpretation and variable binding with assignments, e.g. Heim and Kratzer (1998).

⁹See Meisenbichler (2025) for evidence that the same conclusion applies to the *location coordinate* of an expression’s index, i.e. even though locations can be non-locally bound, their binding is not independent of the expression’s world coordinate.

- b. Nominal predicates exhibit *upper limit effects*, which are patterns of unacceptability that arise because the property time of a predicate is into the future of the predicate's evaluation time.
- c. The distribution of upper limit effects is restricted by the DP's *evaluation world*: if an intensional quantifier α binds the evaluation world of a DP β , then the time that α binds sets the upper limit for the property time of the DP's noun.
- d. To account for c.), we must assume that the evaluation time of a DP is always bound by the same expression that binds that DP's evaluation world.
- e. In a system that employs independent variables for worlds and times, d.) must be stipulated.
- f. The stipulation effectively eliminates any independence between world- and time-variables, which makes a system like (7) superfluous. On the other hand, d.) immediately follows from a system that only employs variables over *evaluation indices*.

2. Upper limit effects beyond complement clauses

2.1. Temporality in DPs

The temporal semantics of DPs is a complex topic in its own right and I won't be able to do it justice here. I will focus on the parts that matter for the purposes at hand and I hope that the overall argument won't be compromised by any shortcuts I have to take (but see Enç, 1981, 1986; Musan, 1995, 1999; Tonhauser, 2021; O'Leary, 2022; among many others, for some of the detail I am glossing over). Here, I want to highlight two points: First, the actual *property time* (PT) of the noun (= the time when the individuals in the noun's extension have the nominal property) is only indirectly related to the nominal *evaluation time* (ET; which is the "variable-like" time coordinate that was non-locally bound in (2) and (5)). Second, the property time can either *overlap* or be *past-shifted* relative to the evaluation time, but PT cannot be *into the future* of ET. Consider examples (10) and (11).

(10) Most days when I worked as a train conductor, *every passenger* was polite.

- (11) a. Scenario: *Over the course of five decades, Mary dedicated herself to an ambitious book project on the psychology of prisoners. She regularly visited prisons to interview inmates. Each time she gained access to a new facility, there were two or three inmates who had previously escaped – individuals with stories of narrow escapes and dramatic accounts of a life on the run that never failed to captivate her. Her research spanned so many years that, on occasion, one of the inmates who had broken out from prison was someone she had interviewed years earlier, long before he had made any attempt to escape. (11b) is from the book:*
- b. Whenever I visited a prison, *every fugitive* told me a captivating story.

In (10), it is only the set of passengers from a given day whose polite behavior counts, which we can capture by having the time variable of *every passenger* bound by the time quantifier *most days*. Importantly, the individuals we are quantifying over do not have to be passengers

throughout the whole day the conductor was at work. Since people get on and off at different stations and at different times throughout the day, the property times most likely do not even overlap across all individuals in the domain.¹⁰ It seems that the nominal property time in this case (= the time of being a passenger) is not identical to the evaluation time that the bound time variable denotes (= the workday of the conductor), but merely *overlaps* the evaluation time span at some point for each individual. We can capture the fact that the property time varies across individuals by assuming that it is introduced by an existential quantifier whose contribution is separate (though not independent) from the time variable.¹¹ In (10), this existential quantifier is restricted by an overlap-requirement to the evaluation time.¹²

- (12) a. $\|passenger\|^g = \lambda t'. \lambda w'. \lambda x. x \text{ is a passenger in } w' \text{ throughout } t'$
 b. $\|PropT_o\|^g = \lambda P_{\langle i, \langle s, \langle e, t \rangle \rangle \rangle}. \lambda t'. \lambda w'. \lambda x. \exists t \circ t' [P(t)(w')(x) = 1]$
 c. $\|PropT_o \text{ passenger}\|^g (\|t_5\|^g) = \lambda w'. \lambda x. \exists t \circ g(5) [x \text{ is a pass. in } w' \text{ through } t]$

(11) shows that overlap is not the only possible temporal relation between PT and ET. Again, we are dealing with a bound variable reading of ET because the relevant (former) fugitives vary with each visit. In this scenario, however, the property time of the relevant individuals must be completely into the past of the evaluation time. At *each* visit time t' , we are considering those individuals at t' that have been a fugitive in the past of t' . Notice that the past-shift of the property time co-varies with the visits (some inmates that are interviewed at two times t' and t'' are not part of the domain of the quantifier at t' but become part of the domain at t'' because their fugitive-time is located in between t' and t''). Also, the property time can again vary from individual to individual (there is no requirement that any two individuals must have escaped or be caught at the same time). To capture these patterns, we can introduce a second silent quantifier that can shift the property time into the past of the evaluation time.¹³

- (13) a. $\|PropT_{<}\|^g = \lambda P_{\langle i, \langle s, \langle e, t \rangle \rangle \rangle}. \lambda t'. \lambda w'. \lambda x. \exists t < t' [P(t)(w')(x) = 1]$
 b. $\|PropT_{<} \text{ fugitive}\|^g (\|t_5\|^g) = \lambda w'. \lambda x. \exists t < g(5) [x \text{ is a fugitive in } w' \text{ through } t]$

Crucially, there is no variant of *PropT* that systematically licenses a *forward-shift* of PT relative to ET. Whereas the past-shift in (11) is readily accepted by most speakers without much difficulty, they typically reject using a sentence like (14) to talk about individuals that would only become fugitives at some point *into the future* of the time when the speaker interviewed them.¹⁴ In other words, even though the evaluation time of a DP does not impose a strict overlap requirement on the nominal property time (i.e. it allows past-shift), it generally imposes an *upper limit* on the left boundary of the property time interval.

¹⁰This observation goes back to Kusumoto (2005).

¹¹Just for the sake of being explicit, I will assume separate tense-like morphemes that introduce the existential property time, but we could have alternatively written it into the lexical entry of the nouns themselves.

¹²Overlap to ET was also the relevant relation that we required to capture non-local binding in (2) and (5).

¹³There are of course several complications that we are ignoring here. For example, the existential past-shift imposes very weak (and most likely *too* weak) requirements on what individuals should be counted as part of the domain. For us it is only relevant that past-shift of any kind is possible whereas future-shift is not. A more adequate account will most likely have to add a more explicit theory of pragmatic relevance and implicit domain restrictions.

¹⁴If such forward-shift is possible at all, then it is very marginal and much harder to access than both overlap or past-shift. To help the intuition, imagine individuals that are interviewed during three visits: once before they escaped and twice after they were back in prison. The relevant reading should ascribe erratic behavior only for the first visit. Note also that we are talking about simple DPs without additional modifiers (e.g. *future*).

- (14) Whenever I visited a prison, *every fugitive* showed signs of erratic behavior.
 (# if the erratic individuals are those that are fugitives in the future of the visits)

2.2. ULEs in complement clauses

Upper limit effects (ULEs) are patterns of unacceptability that are most commonly associated with complement clauses (see Abusch, 1997; Heim, 1994): (15) shows that a verbal complement in the past tense cannot denote a time that is into the future of the attitude time introduced by the embedding verb.

- (15) #When Ruby turned 10 she thought that she was president on her 30th birthday.¹⁵

Note that it is in fact the attitude time (i.e. the self-ascribed time of the attitude holder that *think* quantifies over) and not the matrix tense time that sets the upper limit for the complement clause predicate. We know this because (15) is fine if the attitude holder is mistaken about the actual time: if 10-year-old Ruby somehow believed that she was 30 years old (or older) at the time of her thinking, then the sentence becomes acceptable. We can describe the ULE in complement clauses using the terminology from the last section: the time throughout which the verbal property is true is the verb's *property time*. It makes sense to assume that the property time of a finite verb is determined by tense (perhaps in interaction with aspect). The time that restricts the property time interval by setting an upper limit is the verbal *evaluation time*. Similarly to the property time of nouns, we then find that the property time of a (past-tensed) verbal complement can be either overlapping (namely in *Sequence-of-Tense* contexts) or past-shifted (namely in non-vacuous uses of the past tense) relative to its evaluation time (e.g. Heim, 1994; Ogihara and Sharvit, 2012). Viewed from this perspective, the data in (15) reveal a restriction on how *evaluation times* are selected for complement clauses. If the evaluation time of the complement clause could have been any salient time other than the local attitude time (e.g. the utterance time), then nothing should prevent (15) to express a coherent reading where the embedded property time is into the future of the attitude time. Since this is not the case, the evaluation time of a complement clause seems to be selected based on a strict locality requirement. In other words, the appearance of ULEs in cases like (15) only makes sense if the evaluation time (ET) of a verbal complement clause (in contrast to the ET of DPs) cannot be determined by a non-local binding mechanism (i.e. a mechanism with the expressive power of object language variables). The fact that worlds and times in complement clauses (in contrast to DPs) cannot be non-locally manipulated is part of a constraint known as *Generalization X* (after Percus, 2000, see also Heim, 1991; Farkas, 1994 for earlier observations along these lines). More evidence that this restriction applies in the temporal domain is given in (16): only in the scenario where the complement clause predicate is bound by Susan's internal "now" (the time supplied by the *local* attitude) is the sentence felicitous.

- (16) a. Whenever Susan woke up from a coma, she believed that her 25-year-old student was 30 years old.
 b. DP-time *non-locally* bound, complement predicate *locally* bound
Susan suffers from a rare condition that causes her to fall into a brief coma – lasting about an hour – every few years. She is a college professor, and on the three

¹⁵Imagine the sentence being uttered on Ruby's 35th birthday.

Upper limit effects in DPs and the interdependence of worlds and times

occasions she experienced such comas (in 1990, 1996, and 2001), her then 25-year-old graduate student was with her and drove her to the hospital. Strikingly, each time she awoke, Susan insisted that she had been asleep for exactly five years. When questioned in 1990, she claimed it was 1995 and that her graduate student must be 30 years old. In 1996, she insisted it was 2001, once again asserting that her accompanying student was 30.

- c. # DP-time locally bound, complement predicate non-locally bound
Susan suffers from a rare condition that causes her to fall into a brief coma – lasting about an hour – every few years. She is a college professor, and on the three occasions she experienced such comas (in 1990, 1996, and 2001), her then 30-year-old graduate student was with her and drove her to the hospital. Strikingly, each time she awoke, Susan insisted that she was located five years in the past from the time the coma set in. When questioned in 1990, she claimed it was 1985 and that her graduate student must be 25 years old. In 1996, she insisted it was 1991, once again asserting that her accompanying student was 25.

2.3. ULEs in DPs and why they are puzzling

We have concluded that the robust appearance of ULEs that attitude times impose on verbal complements follows from the fact that the attitude time *must* be selected as the complement's evaluation time (due to the strict locality requirement *Generalization X*). We have also seen that the evaluation time of DPs is *not* subject to such a locality requirement: even if the DP takes scope inside the attitude context, any higher temporal quantifier can in principle bind its evaluation time. Based on this, a reasonable expectation is that DPs will have an easier time escaping any upper limit violations. Although this is mostly true, there are a few surprising cases where DPs very robustly trigger ULEs. Consider (17) and (18).

- (17) a. Scenario: *Erik works as a professional psychic. In the year 2000, he is hired to perform at Amy's tenth birthday party. He asks each child attending the party to draw a card from a deck, which contains four identical copies of each card. Without seeing which child drew which card, Erik makes several predictions: the children who drew the Queen of Spades would become millionaires by 2015; those who drew the King of Diamonds would become senators; and those who drew the Ten of Hearts would be in prison for murder by 2020. Erik sincerely believed in these predictions, but he turned out to be completely wrong. Every child at the party grew up to be an upstanding citizen, and none of them ever went to prison.*
- b. # When Erik performed a show for Amy's 10th birthday, he believed that every prisoner drew a 10 of Hearts.
- (18) a. Scenario: *In the year 2000, Anne and her nephew Declan visit Bryan and his eight children. At some point during the visit, Anne realizes that her wallet is missing. She immediately becomes suspicious of Bryan's children. "I always knew that some of them would turn into criminals" she says. "In fact, I'm making a prediction: twenty years from now, whichever child stole my wallet – whoever it may be – will be in prison for murder." Her prediction turns out to be wrong. All of Bryan's children grow up to be upstanding citizens. In 2020, Declan admits:*

“Anne didn’t realize that it was me who stole her wallet. To her, I was an angel and only children with criminal potential were suspects. (18b)”

- b. # In fact, when we were at Bryan’s house 20 years ago, Anne believed that a murderer stole her wallet.

The unacceptability of these examples seems to arise because the property times of *every prisoner* and *a murderer* are into the future of their respective attitude time.¹⁶ But if the evaluation time sets the upper limit for the property time and the evaluation time of a DP can be non-locally bound, why can these DPs not take the utterance time as their evaluation time, which should make (17) and (18) acceptable in these scenarios (since $PT < / \circ$ utterance time)?

(19) *Why is this not a possible LF for (17)?*

LF: $\lambda t_0 \dots$ believe $\lambda t_1 \dots$ [every t_0 $PropT_{</\circ}$ prisoner]

The answer to (19) seems to have something to do with the fact that the DPs in (17) and (18) are *opaque*, i.e. their *evaluation world* is locally bound by the attitude verb.

2.4. Evaluation worlds restrict the distribution of ULEs

Even though opaque readings imply narrow scope with respect to the attitude verb, non-local binding data like (2) already show that scope does not constrain the selection of a DP’s evaluation time. This means scope cannot be the deciding factor behind the unacceptability of (17) and (18). This leaves *opacity* (i.e. world binding) itself as the potential reason that the utterance time cannot be taken as an antecedent for the evaluation time. Indeed, once we control for scope *and* world binding, the generalization becomes clear: in (20), the DP *every tenured professor in our department* scopes inside the attitude context (controlled for by narrow scope relative to the negative existential), but its evaluation world is the actual world. Anton has no idea that Axel, Andrew, Mona, and Mina would all become professors in the psychology department (based on his negative opinion of them, he would most likely deny this possibility). Note now two things about (20): a.) the evaluation time of the DP is the utterance time, which is properly into the future of the attitude time, b.) the target sentence is nevertheless acceptable.

- (20) a. Scenario: *There are exactly four tenured professors in the psychology department where Charlotte is doing her post-doc: Axel, Andrew, Mona, and Mina. Eight years ago – several years before they became professors – a rumor circulated that*

¹⁶See O’Leary (2022) and Meisenbichler (2023) for some discussion of similar judgments and what the relevant generalization might be. O’Leary concludes that data such as (18) show us that DPs in the scope of an attitude verb cannot take the *utterance time* (specifically) as their property time. According to O’Leary, this is due to a local binding requirement on evaluation times of the type we found for complement clauses. This conclusion – according to which the evaluation time is determined in a DP’s scope position by the closest c-commanding binder – is a non-starter given data like (2) and (5), which clearly show that there is a mechanism for binding evaluation times of DPs non-locally. Moreover, it can be shown that the unacceptability of (18) is not about the utterance time *per se*, but about a forward-shift more generally. This is possible because we independently know that the property time of a DP does not have to match a time that is introduced in the sentence, but can also be a contextual time (e.g. Enç, 1981). Imagine a scenario in which Pierre’s dog gets killed in the year 1805 and (because of a vision he had) he is absolutely certain that he would catch the killer and put him in prison. In fact, however, the actual killer of Pierre’s dog never gets caught. A historian in the year 2020 cannot felicitously say: “#After he discovered his dead dog in 1805, Pierre thought that it was killed by a prisoner.”

Upper limit effects in DPs and the interdependence of worlds and times

all four of them were having an affair with an influential millionaire. Now some people even say that it was the influence of that millionaire that got them their job at the university. When discussing this with a colleague, Charlotte recounts an occasion eight years earlier when all four of them were attending a party at Anton's house, at a time when they were all still students. Charlotte remembers telling Anton about the rumor. Having formed the opinion that the four of them are unpleasant and arrogant, Anton quickly dismissed it: "It is unimaginable to me that there would be a person with such bad taste. Maybe there is a millionaire who is dating one or two of them, but all four? That's simply absurd". Charlotte reports what Anton thought about the rumor by uttering (20b).

- b. (I remember that at his party eight years ago) Anton firmly believed that no millionaire was having an affair with *every tenured professor in our department*.

We can state the following generalization about the distribution of upper limit effects:

- (21) World-dependent distribution of upper limit effects
If α is a quantifier over worlds and times, and α binds the evaluation world of β , then the time quantified over by α sets the upper limit for the property time of β .

Another way of saying this: if an intensional quantifier binds the evaluation world of an expression, then it also binds its evaluation time. To confirm that this is the correct way to think about these data, let's probe our intuitions by looking at an example with multiple embeddings. Read (22) as a continuation of (17): twenty years after Amy's birthday, she is trying to remember what the psychic Eric predicted on the day of her birthday party. Let's consider different versions of Amy's beliefs that vary the modal and temporal interpretation of the DP *every prisoner*. What we find in (23) and (24) below supports the conclusion that binding of a DP's evaluation time is always done by the antecedent of the DP's evaluation world.

- (22) Twenty years after her birthday party, Amy believed that Erik had claimed that *every prisoner* (at the party) had drawn a Queen of Spades.

- (23) Cases where world-dependent ULE (21) is violated are unacceptable

- a. ✗ *claim* binds evaluation world (EW), property time (PT) > *claim-time*:
In 2020, Amy remembered that Erik had predicted at the party (in 2000) that the children who drew the Queen of Spades would end up as prisoners by 2010. However, now, looking back, Amy knew that prediction had been false.
- b. ✗ *believe* binds EW, PT > the attitude time of *believe*:
In 2020, Amy was convinced that Jack, Tom, and Bill were criminals. Although they hadn't been caught yet, she was certain it was only a matter of time before they ended up behind bars. In fact, she was certain they'd all be in prison by 2024. It turns out that Jack, Tom, and Bill would never be arrested: they die in a car accident in 2025, long before their criminal activities could be discovered (and before the utterance time). Amy also recalled Erik's prediction at the party in 2000, where he had guessed that Jack, Tom, and Bill had drawn the Queen of Spades. According to Amy, Erik had gone on to predict that they would become upstanding citizens.

- c. ✗ utterance world binds EW, PT > utterance time:
In the future, Jack, Tom, and Bill will be arrested and sentenced to life in prison, a fate Amy would have never guessed for them in 2020. She saw them as incapable of any criminal behavior. Back then, Amy also remembered Erik's prediction at the party in 2000, where he had guessed that Jack, Tom, and Bill had drawn the Queen of Spades. She recalled him saying that they would go on to become upstanding citizens who would never do anything criminal all their life.

(24) Cases where world-dependent ULE (21) is obeyed are acceptable

- a. ✓ claim binds EW, PT < the attitude time of claim:
In 2020, Amy believed Erik was insane. She recalled that, at the party in 2000, he had claimed some of the people who participated in the card-drawing had been imprisoned in the early '90s and that every one of those former prisoners had drawn the Queen of Spades.
- b. ✓ believe binds EW, PT < the attitude time of believe:
In 2020, Amy believed that Jack, Tom, and Bill had been in prison sometime between 2007 and 2014. She was mistaken, however. In reality, Jack, Tom, and Bill had been upstanding citizens all along, and the story of their imprisonment had been nothing more than a false rumor. Amy also recalled Erik's prediction at the party in 2000, where he had guessed that Jack, Tom, and Bill had drawn the Queen of Spades. She remembered him saying they would go on to become upstanding citizens who would never do anything criminal all their life.
- c. ✓ utterance world binds EW, PT < the utterance time
Jack, Tom, and Bill were in prison for a burglary from 2022 to 2025. In 2020, Amy would never have imagined such a fate for them. She saw them as upstanding citizens, incapable of being criminals. At that time, Amy also recalled Erik's prediction from the party in 2000, where he had guessed that Jack, Tom, and Bill had drawn the Queen of Spades. She remembered Erik saying they would go on to become upstanding citizens who would never do anything criminal all their life.

2.5. Coordinate variables vs. index variables

The distribution of ULEs in DPs points to the following restriction:

- (25) The evaluation time of an expression α has to be from the same source as α 's evaluation world.

The point now is the following: if the world- and time-coordinate of a DP are represented as *independent* variables in the object language (e.g. (7)), there is no straightforward way to capture the restriction in (25) without stipulation.¹⁷ In such a system, we can quite naturally state LFs that derive exactly the “mixed” cases of binding that (25) is trying to rule out.

- (26) *Why is this not a possible LF for (17)?*
 a. LF: $\lambda t_0 \lambda w_0 \dots \text{believe } \lambda t_1 \lambda w_1 \dots [\text{every } w_1 t_0 \text{ Prop}T_0 \text{ prisoner}]$

¹⁷The same of course applies to any other mechanism with a similar expressive power.

- b. $\|PropT_o \text{ prisoner}\|^g(\|t_0\|^g)(\|w_1\|^g) = \lambda x. \exists t \circ g(0) [x \text{ is a prisoner in } g(1) \text{ at } t]$

The only way to capture (25) in this system is *by stipulation*. Even worse, adding a stipulation like (25) makes having two independent mechanisms of non-local variable binding – one for worlds and one for times – completely pointless. If world- and time-binding can never be dissociated (which is what (25) effectively says), then there can never be empirical evidence for their (independent) existence. A reasonable conclusion to draw from this is that world- and time-binding *do not* actually access two independent, dissociated mechanisms. In other words, natural language might employ a mechanism that has the expressive power of *object language index variables* (variables over world-time pairs), but no mechanism that targets the world- or the time-coordinate of the index as if they were independent variables. As soon as we only allow an intensional system with variables that range over indices (world-time pairs), a condition like (25) falls out automatically.¹⁸

(27) *Variables over world-time pairs: no indexing for (17) that allows the illicit reading*

- a. LF: $\lambda i_0 \dots \text{believe } \lambda i_1 \dots [\text{every } i_0 \text{ } PropT_o \text{ prisoner}]$
- b. $\|PropT_o \text{ prisoner}\|^g(\|i_0\|^g) = \lambda x. \exists t \circ T(g(0)) [x \text{ is a prisoner in } W(g(0)) \text{ at } t]$
- c. LF: $\lambda i_0 \dots \text{believe } \lambda i_1 \dots [\text{every } i_1 \text{ } PropT_o \text{ prisoner}]$
- d. $\|PropT_o \text{ prisoner}\|^g(\|i_1\|^g) = \lambda x. \exists t \circ T(g(1)) [x \text{ is a prisoner in } W(g(1)) \text{ at } t]$

3. Other cases of non-local index binding

3.1. Degree clauses

DPs are not the only environments that allow non-local binding of their evaluation world (and time). Famously, comparative clauses can receive transparent readings even if they take scope below an attitude verb or inside the antecedent of a conditional (see Hoeksema, 1984; Heim, 1985, 2000; Kennedy, 1995). This suggests that world- and time-binding in degree clauses makes use of the same (non-local) mechanism that binds the world- and time-coordinate of DPs. In contrast to DPs, however, degree clauses can host overt tenses, which gives us a new combination to test: a tensed predicate whose evaluation is not subject to Generalization X. If the conclusions from previous sections are correct, we expect a tense in degree clauses to pattern with DPs when it comes to the distribution of ULEs: if the degree clause is opaque, the local attitude time should impose an upper limit on the tense denotation; if the degree clause is (narrow scope) transparent, a relative forward-shift of the degree clause tense with respect to the attitude time should be licensed. This prediction is correct:

- (28) In the year 2000, Alice believed that none of the buildings in town was higher *than the king's palace was high*.
- a. ✓ Transparent & PT > attitude time
The king's palace didn't exist until 2008, and it was demolished after the revolution in 2015. The palace was massive, with exactly 50 floors. In 2000, Alice made a bet with a friend about the maximum height of the buildings in town. At that time, Alice (wrongly) guessed that no building in town had more than 50 floors.

¹⁸Read $T(i')$ as “the time coordinate of the index i' ” and $W(i')$ as “the world coordinate of the index i' ”.

b. ✗ Opaque & PT > attitude time

In 2000, rumors spread that the king was building a grand palace for himself. Alice believed the rumors and became convinced that it would be the tallest building in town. She assumed that the largest building in 2000 had around 47 floors and that the king's palace would have at least 60. However, the palace plans were never realized, as the monarchy was overthrown just a few years later.

(29) At the beginning of 2001, the investor expected at least one of his companies to lose less money that year *than it lost in 2005*.

a. ✓ Narrow scope+transparent & PT > attitude time

In 2001, Bob invests in three start-ups. At the start of 2001, he sets specific goals for each company. For Company A, the goal is to lose less than \$1 million in 2001. For Company B, the target is to lose less than \$2 million that year, and for Company C, the goal is to limit losses to under \$3 million for 2001. While Bob doesn't expect all these targets to be met, he believes at least one of the companies should achieve its goal. The goals turned out to be very unrealistic and each company lost over 10 million in their first year. To illustrate how bad Bob's estimates were: even in year four (2005), when Bob was already expecting profits, the losses were still higher than what the investor had predicted for 2001: Company A lost slightly over \$1 million in 2005, Company B \$2 million, and Company C \$3 million.

b. ✗ Narrow scope+opaque & PT > attitude time

In 2001, Bob invests in three start-ups. At the beginning of the year, he sets a four-year goal for his investments: by 2005, he expects at least one of his companies to grow double in size. He expects that this will mean an increase in costs (and ultimately losses) for the first few years of this investment. He doesn't expect losses to decrease until year six. For example, he estimates losses of about \$1 million for 2001 and losses of about \$1.5 million in 2005 for the growing business. He turns out to be wrong: all three companies turn a profit by the end of the third year.

These examples show that a past tense in a degree clause (in contrast to complement clause tenses) can in principle denote a time that is into the future of the time that the most local time binder introduces. In (28a), the palace does not even exist at the time of Alice's belief. Since we are talking about the palace's actual height, this "forward-shift" involves a transparent reading. The same is true with (29a), where we additionally control for narrow scope by making sure that a quantifier (*at least one of his companies*) binds a pronoun inside the degree clause (*it*) from within the belief-context. Crucially, as soon as we make sure that the evaluation world of the comparative clause is locally bound by the attitude verb (resulting in an opaque reading), forward-shift relative to the attitude time yields unacceptability (see (28b) and (29b)). What we seem to find is exactly the pattern we found with DPs: evaluation time selection (and therefore the distribution of ULEs) is completely predictable based on the selected evaluation world.

3.2. Relative clauses

Entirely in line with the observations we made about worlds and times, Kusumoto (2005) provides evidence that ULEs in relative clause tenses are restricted by *opacity*, not scope. Specifically, Kusumoto shows that a relative clause in the past tense can denote a time that is into the

future of the time bound by an intensional operator even if it appears in its local scope. For example, the DP in (30a) needs to be interpreted low for NPI-licensing, but the relative clause predicate is true at a time that is into the future of the matrix tense time. Moreover, this “future-shifted” reading of the relative clause past is only possible if the relative clause is *transparent* with respect to the intensional operator (i.e. if the operator doesn’t bind its evaluation world): (30b) is not acceptable if the relative clause description was part of the content of Tom’s claim. Even if Tom had a premonition that Mary, Barbara and Sue will win the Nobel Prize in 2025, (30b) cannot express his disappointment that none of the professors wanted to accept these future Nobel laureates to the grad program. As with our examples for simple DPs, the problem seems to be that an opaque reading of this sort cannot be combined with a future-shift relative to the attitude time. On the other hand, (30b) does have an acceptable reading in which the relative clause description is transparent (i.e. its evaluation world – and thereby its evaluation time – is non-locally bound). This is the case if e.g. Gina, Nora and Magda in actual fact won the Nobel Prize in 2025 and the sentence reports Tom’s claim that, back when the three were applying to grad school, no professor at the department wanted to accept any of them.

- (30) a. She failed to talk to any prospective student *who decided to come to MIT*.¹⁹
b. In our first year of grad school (in 2010), Tom claimed that no professor that year wanted to accept *anyone who won the Nobel Prize of 2025*.

4. Conclusion

It is a well-established fact that natural language has a mechanism that can manipulate world- and time-arguments *non-locally*. A mechanism that can establish non-local dependencies of this sort could in principle treat worlds and times independently, e.g. if worlds and times can be both represented as indexed variables in the object language. In this paper, I argued that this is not the case: worlds and times do not behave like independent variables when they are targeted for non-local binding; they always must be bound by the same antecedent, suggesting that the binding mechanism treats them as a unit (e.g. as an object language variable that ranges over *indices* or *world-time pairs*). If the evaluation world and the evaluation time of an expression α could be selectively targeted by distinct intensional operators, then the availability of non-local binding should allow us to circumvent *upper limit effects* (ULEs): the evaluation time of α could be bound by a non-local antecedent that quantifies over times that are into the future of the times that the antecedent of α ’s world coordinate otherwise sets as an “upper limit”. The fact that environments that allow non-local binding of their world/time-coordinate (DPs, degree clauses, relative clauses) still display ULEs cannot be explained if the non-local mechanism can selectively target worlds and times. Any stipulation that could be added to such a mechanism to account for the distribution of ULEs in these environments would effectively weaken it to something that is equivalent to *index binding*.

References

Abusch, D. (1994). The scope of indefinites. *Natural Language Semantics* 2(2), 83–136.

¹⁹See Kusumoto (2005: 327) for this example and Kusumoto’s paper more generally for more examples.

- Abusch, D. (1997). Sequence of tense and temporal de re. *Linguistics and Philosophy* 20(1), 1–50.
- Bäuerle, R. (1983). Pragmatisch-semantische Aspekte der NP-Interpretation. In M. Faust, R. Harweg, W. Lehfeldt and G. Wienold (Eds.), *Allgemeine Sprachwissenschaft, Sprachtypologie und Textlinguistik: Festschrift für Peter Hartmann*, pp. 121–131. Tübingen: Narr.
- Cresswell, M. (1990). *Entities and indices*. Boston: Kluwer.
- Elbourne, P. (2013). *Definite descriptions*. Oxford: Oxford University Press.
- Elliott, P. (2023). A flexible scope theory of intensionality. *Linguistics and Philosophy* 46(2), 333–378.
- Enç, M. (1981). Tense without scope: An analysis of nouns as indexicals. Ph.D. thesis, University of Wisconsin, Madison.
- Enç, M. (1986). Towards a referential analysis of temporal expressions. *Linguistics and Philosophy* 9(4), 405–426.
- Farkas, D. (1994). An indexical theory of scope. *Acta Linguistica Hungarica* 42(3/4), 283–303.
- Fodor, J. D. (1970). The linguistic description of opaque contexts. Ph.D. thesis, Massachusetts Institute of Technology.
- Heim, I. (1985). Notes on comparatives and related matters. Manuscript, University of Texas, Austin.
- Heim, I. (1991). Artikel und Definitheit. In A. von Stechow and D. Wunderlich (Eds.), *Semantik: Ein internationales Handbuch der zeitgenössischen Forschung*, pp. 487–535. Berlin: De Gruyter.
- Heim, I. (1994). Comments on Abusch’s theory of tense. In H. Kamp (Ed.), *Ellipsis, tense and questions*, pp. 143–170. Edinburgh: University of Edinburgh.
- Heim, I. (2000). Degree operators and scope. In B. Jackson and T. Matthews (Eds.), *Proceedings of SALT 10*, pp. 40–64. Ithaca NY: CLC Publications.
- Heim, I. and A. Kratzer (1998). *Semantics in Generative Grammar*. Oxford: Blackwell.
- Hoeksema, J. (1984). To be continued: The story of the comparative. *Journal of Semantics* 3(1–2), 93–107.
- Kamp, H. (1971). Formal properties of ‘now’. *Theoria* 32(3), 227–273.
- Kennedy, C. (1995). Comparatives, indices, and scope. *Papers from the Sixth Annual Meeting of the Formal Linguistics Society of Midame*, 115–126.
- Kusumoto, K. (2005). On the quantification over times in natural language. *Natural Language Semantics* 13(4), 317–357.
- Meisenbichler, P. (2023). On the interaction of worlds and times. MA thesis, University of Vienna.
- Meisenbichler, P. (2025). Interactions of worlds, times and locations: The limits of (non-local) index shifting. In D. Demiray, R. Cheng-yen Liu and N. Segal (Eds.), *Proceedings of the 55th Annual Meeting of the North East Linguistic Society (NELS 55)*, pp. 137–147. Amherst: GLSA Publications.
- Montague, R. (1973). The proper treatment of quantification in ordinary English. In J. Hintikka, J. Moravcsik and P. Suppes (Eds.), *Approaches to natural language*, pp. 221–242. Reidel.
- Musan, R. (1995). On the temporal interpretation of noun phrases. Ph.D. thesis, Massachusetts Institute of Technology.
- Musan, R. (1999). Temporal interpretation and information-status of noun phrases. *Linguistics and Philosophy* 22(6), 621–661.
- Ogihara, T. and Y. Sharvit (2012). Embedded tenses. In R. Binnick (Ed.), *Handbook of Tense and Aspect*, pp. 638–668. Oxford University Press.

Upper limit effects in DPs and the interdependence of worlds and times

- O’Leary, M. (2022). About time: Lexical, structural, and discourse constraints on the temporal interpretation of nominal predicates. Ph.D. thesis, University of California, Los Angeles.
- Percus, O. (2000). Constraints on some other variables in syntax. *Natural Language Semantics* 8(3), 173–229.
- Schwarz, F. (2012). Situation pronouns in determiner phrases. *Natural Language Semantics* 20(4), 431–475.
- Tonhauser, J. (2021). Temporal properties of noun phrases. In D. Gutzmann, L. Matthewson, C. Meier, H. Rullmann and T. Zimmermann (Eds.), *The Wiley Blackwell Companion to Semantics*. Hoboken, NJ: John Wiley and Sons.
- Vlach, F. (1973). “Now” and “Then”: A formal study in the logic of tense anaphora. Ph.D. thesis, University of California, Los Angeles.
- von Stechow, K. and I. Heim (2011). Intensional semantics. MIT lecture notes.
- von Stechow, A. (2004). Binding by verbs: Tense, person, and mood under attitudes. In H. Lohnstein and S. Trissler (Eds.), *The Syntax and Semantics of the Left Periphery*, pp. 431–488. Berlin/Boston: De Gruyter.